




GEORGE STEVE FAJARDO SORIA

ABOUT ME

Research assistant with a background in aerospace and experience designing and implementing machine/deep learning algorithms and control theory.

SKILLS:

Technical: Python, C++, Github, STK certification, Matlab, Solidworks.
Soft: Resilience, Team working.



WORK EXPERIENCE

Research assistant

Peruvian Space Agency (CONIDA) | Dec 2022 – Present

My primary role is the **mission design** of the **"KILLASAT" cubesat** and **developing and implementing** the Attitude Determination and Control Subsystem (**ADCS**), **including component selection, simulation in MATLAB and microcontroller programming.**

In addition, I support the planning, implementation, and management of a program to promote aerospace development in the space agency and the country's research centers.

Undergraduate Teacher

Toulouse Lautrec School of Higher Education (Visiva) | Dec 2022 - Present

I guide students in advanced tech projects, teach website creation and innovation plans, facilitate 3D printing, 3D design, Arduino/ESP32 and programming, and lead training in high-definition prototyping and product experimentation.

Research assistant

Jicamarca Radio Observatory (JRO) | Jan 2020 – Nov 2022

I have developed several **machine learning and deep learning algorithms (python)** to correlate ionospheric phenomena and their impact on Global Navigation Satellite System (GNSS) receivers in a network located all over South America. Improved ionospheric scintillation (S4) forecast accuracy metrics over the literature.

Chief Coordinator of the laboratory

Chief Coordinator of the laboratory | Dec 2018 – Dec 2019

Five aerospace engineering training projects were managed—approximately 40 undergraduate students. **Two successful launch campaigns were achieved.** A new work methodology was proposed to increase scientific publications.



EDUCATION

Universidad Nacional de Ingeniería, Lima

Bachelor of Engineering, Mechatronics | August 2014 - December 2019 (**5 years studies program**)

Universidad Nacional Autónoma de México, Mexico City

Undergraduate Exchange Student | August 2017 - December 2017

Diploma in Mechatronics Systems, FESTO

FESTO, October 2017 - January 2018



RESEARCH

"The impact of the Tonga volcanic eruption on the Peruvian atmosphere: from the sea surface to the ionosphere"

Co-author, analyzing GPS receivers signals for a paper under revision in Journal "Earth, Planets and Space"-Dec 2023

Initiatives of the Peruvian Space Agency (CONIDA) to boost the growth of the Peruvian aerospace ecosystem

Main author in paper published in International Astronautical Congress (IAC). Code: [IAC-23,B4,1,3,x80175](#) | Oct 2023

Lessons learned from Smart Machines - CITC UNI. Program for Capacity Building in Peruvian Undergraduates and perspectives in the Near Future

Main author in paper published in International Astronautical Congress (IAC). Code: [IAC-21.E1.3.13](#) | Mar 2021 - Sep 2021

Conceptual Design of a Mars Constellation for Global Communication Services using Small Satellites

Coauthor in paper published in International Astronautical Congress (IAC). Code: [IAC-20.D3.2A.2](#). Space Generation Advisor Council (SGAC). | Jan 2020 - Sep 2020

WANKA stratospheric balloon Modeling and control of the fall of a payload through the use of a low cost parachute - WANKA stratospheric balloon

Sep 2018 - December 2022

Paper: Wanka - A mission to measure stratospheric aerosols concentration using low-cost commercial sensors onboard a high-altitude balloon - 73rd International Astronautical Congress 2022. [IAC-22,E2,3-GTS.4,15,x71364](#)

Ionospheric effects study in the signs of radiofrequency through analysis of data GNSS receivers by using machine learning techniques.

Advisor: Dr. Pacheco Edgardo. Posters presented at CEDAR Virtual Workshop (2020,2021), | Jul 2019 - Dec 2021

Development of telemetry using Machine learning for Rover participant in Human Exploration Rover Challenge (HERC) NASA

Student Section ASME, National University of Engineering.

Telemetry Award Human Exploration Rover Challenge NASA. | Nov 2018 - Apr 2019

I developed an algorithm to track rovers without GPS, using a machine learning algorithm,



CONFERENCES AND COURSES

30th Workshop on Space Technology for Socio-Economic Benefits: "Challenges and Capacity-building Opportunities for Emerging Space Nations" - UNOOSA / IAC

Panelist funded by UNOOSA: Talking about the development of the peruvian aerospace ecosystem | Oct 2023.

United Nations Workshop on the International Space Weather Initiative: The Way Forward

Deegate fully funded by UNOOSA. Presenting findings in the GPS receivers of LISN. Impacts of space weather in GPS receivers | Jul 2023

International Workshop on Machine Learning for Space Weather: Fundamentals, Tools and Future Prospects

International Centre for Theoretical Physics (ICTP) | Jul 2022

Latam Regional Workshop on SciTinyML: Scientific Use of Machine Learning on Low-Power Devices

International Centre for Theoretical Physics (ICTP) | Jul 2022

South American Space Generation Workshop (Online Delegate)

Virtual Delegate | Apr 2021, May 2022.

NASA Space Apps Challenge 2021

Hackaton organized by NASA. Our project was "Universal nominated" | Oct 2021

Eastern Africa GNSS and Space Weather Capacity Building Workshop

International Centre for Theoretical Physics (ICTP) | Jun 2021

Spring School ICTP Complex Systems

International Centre for Theoretical Physics (ICTP) | Mar 2021

Program for Scientific Initiation (Machine Learning)

The National Institute for Telecommunications Research and Training -UNI (INICTEL-UNI) | Jan-Feb 2021

Space Generation Congress (SGC)

Virtual Delegate | Aug 2020

Peruvian Mission to Mars (Misión Peruana a Marte)

Hackathon where the necessary solutions were developed in a hypothetical situation in which Peru established a Martian colony. | May 2020

5th South American Space Generation Workshop

Escuela Militar de Ingeniería, Bolivia. Oct 2019

The Third Summer Camp of APSCO Student Small Satellite Project

Shangai Jiao Tong University (SJTU), China. Aug 2019

International Conference on CubeSat Technology

Peruvian Space Agency, Peru and Brazilian Space Agency, Brazil. Oct 2018

XIV International Summer Space School

Samara State Aerospace University, Russia. Aug 2018



EXTRA-CIRRUCULAR

Journal Club Aerospace

A space where students were taught to analyze scientific research. Each meeting included presentations and discussions of previously studied research. | Oct 2020-Jan2021

AIDA-UNI (Aerospace Research and Development Association)

Co-Founder

AIDA was founded to help school students enroll in STEM careers and help freshman and sophomre university students obtain research experience in our university and abroad. | Aug 2019-Present